

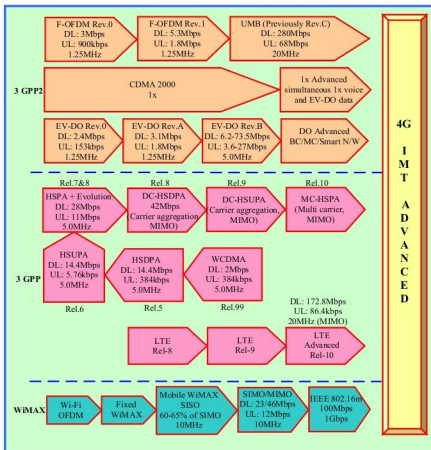


Fig. 1: 4G candidate technologies

flatter network compared to HSPA +, but round trip delays ranging between 100ms and 140ms put WiMAX quite far from IMT-Advanced targets.

Even though none of these technologies are fulfilling the IMT-Advanced criterion, each one has an evolution path to get there. An evolution path to 4G is depicted in Fig. 1.

HSPA + is being expanded to allow 40MHz channelisation with 4x4 MIMO capabilities that allows it to achieve 16.8bit/s/Hz peak spectral efficiency. Details about how to achieve other targets are all part of release 11, which is expected to be completed very soon.

LTE is closer to 4G thresholds compared to HSPA+ and was formally submitted by the 3GPP organisation. Primarily, LTE was designed to incorporate MIMO from the start. If 4x4 MIMO is implemented in the terminal form factor to have an LTE-category five devices, then using

LTE-Advanced that allows 40MHz to 100MHz channelisation will bring true 4G capabilities.

LTE-Advanced is part of 3GPP release 10 and is essentially an enhancement to the existing LTE network. This upgrade path makes it more cost-effective for vendors to offer LTE and then upgrade to LTE-Advanced. LTE-Advanced (release 10) has been formally submitted by 3GPP in fall 2009 to achieve high data rates of 1Gbps peak download and 500Mbps peak upload, and it is expected to be released soon.

On WiMAX front, IEEE 802.16m or wireless MAN-Advanced evolution of 802.16e (mobile WiMAX) is under development, with the objective to fulfil the IMT-Advanced criteria of 1Gbps for stationary reception and 100Mbps for mobile reception.

Some other technologies like 1x-Advanced, DO-Advanced and ultra mobile broadband (UMB) are

also rapidly progressing to meet and surpass the IMT-Advanced requirements. Following a detailed evaluation against stringent technical and operational criteria, ITU has determined that LTE-Advanced and IEEE 802.16m/wireless MAN-Advanced should be accorded the official designation of IMT-Advanced, which is considered as 4G.

In a nutshell

The wonder technology 4G is expected to cover the data-rate deficiency in previous generations. It is also aimed to achieve quality of services including voice chat, video calls, MMS, HDTV and voice-over Internet protocol (VoIP). This technology aims to provide 100Mbps data rate for moving users and 1Gbps for stationary users so that they can enjoy uninterrupted connectivity with high speed anywhere they go.

Flawless connectivity and international roaming across innumerable networks of the world is its next target. It is an all-IP-switched network and several working groups propose that it should offer an open Internet protocol. The technology components being considered for 4G include carrier aggregation, advanced access schemes, enhanced multiple antenna transmission, relaying and repeater solutions, among others. In addition to these solutions to enhance coverage and cell edge data rates, an evolution of inter-cell interference coordination in the form of coordinated multi-point transmission/reception is yet another technology to enhance the 4G network performance.

4G candidate technologies are LTE-Advanced, wireless MAN-Advanced or mobile WiMAX 802.16m, 1x-Advanced, UMB, DO Advanced and MC-HSPA. Recently, ITU has determined that LTE-Advanced and wireless MAN-Advanced should be accorded the official designation of 4G. Capabilities of IMT-Advanced/4G systems are being continuously enhanced in line with user trends and technology developments. ●